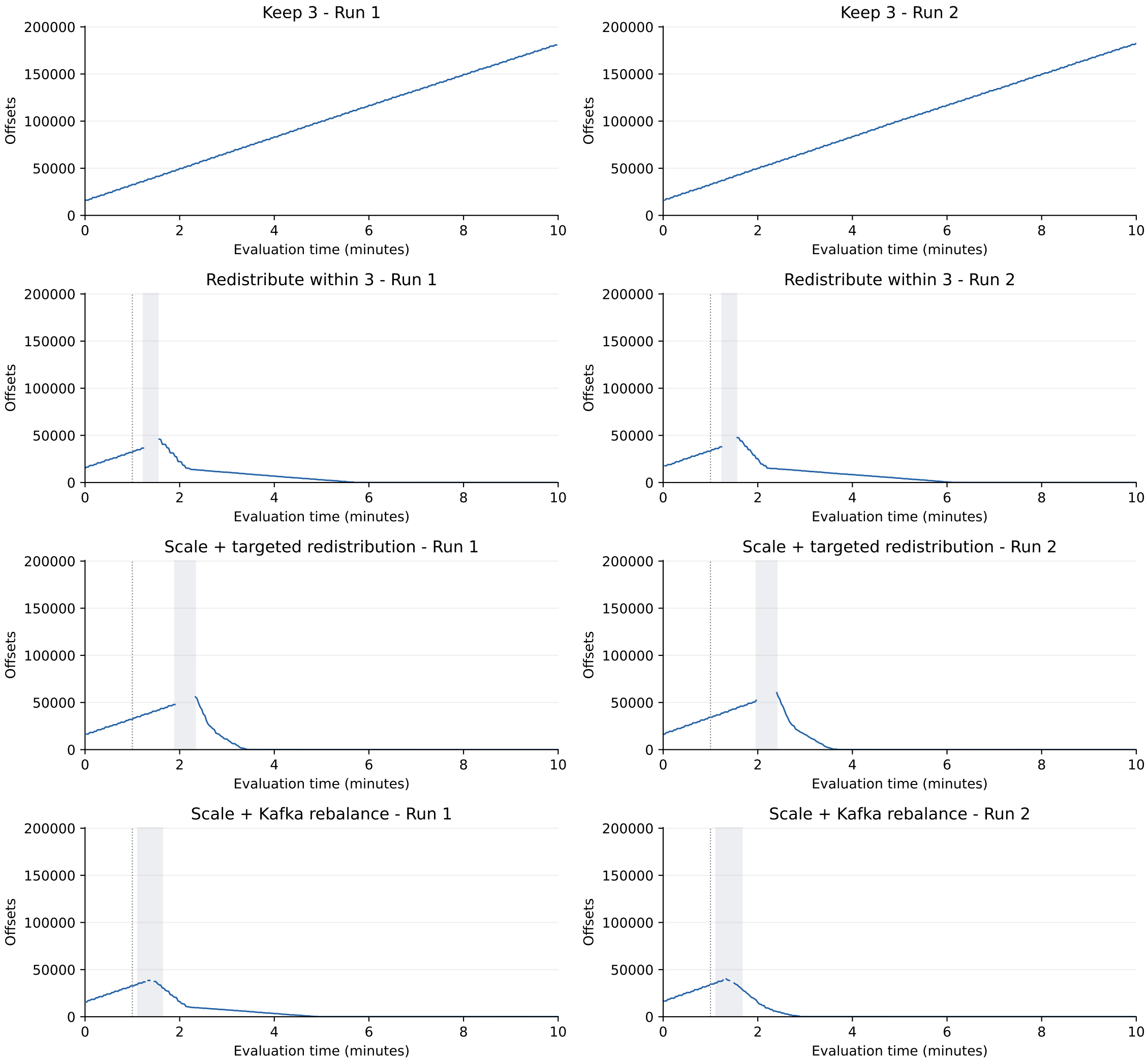
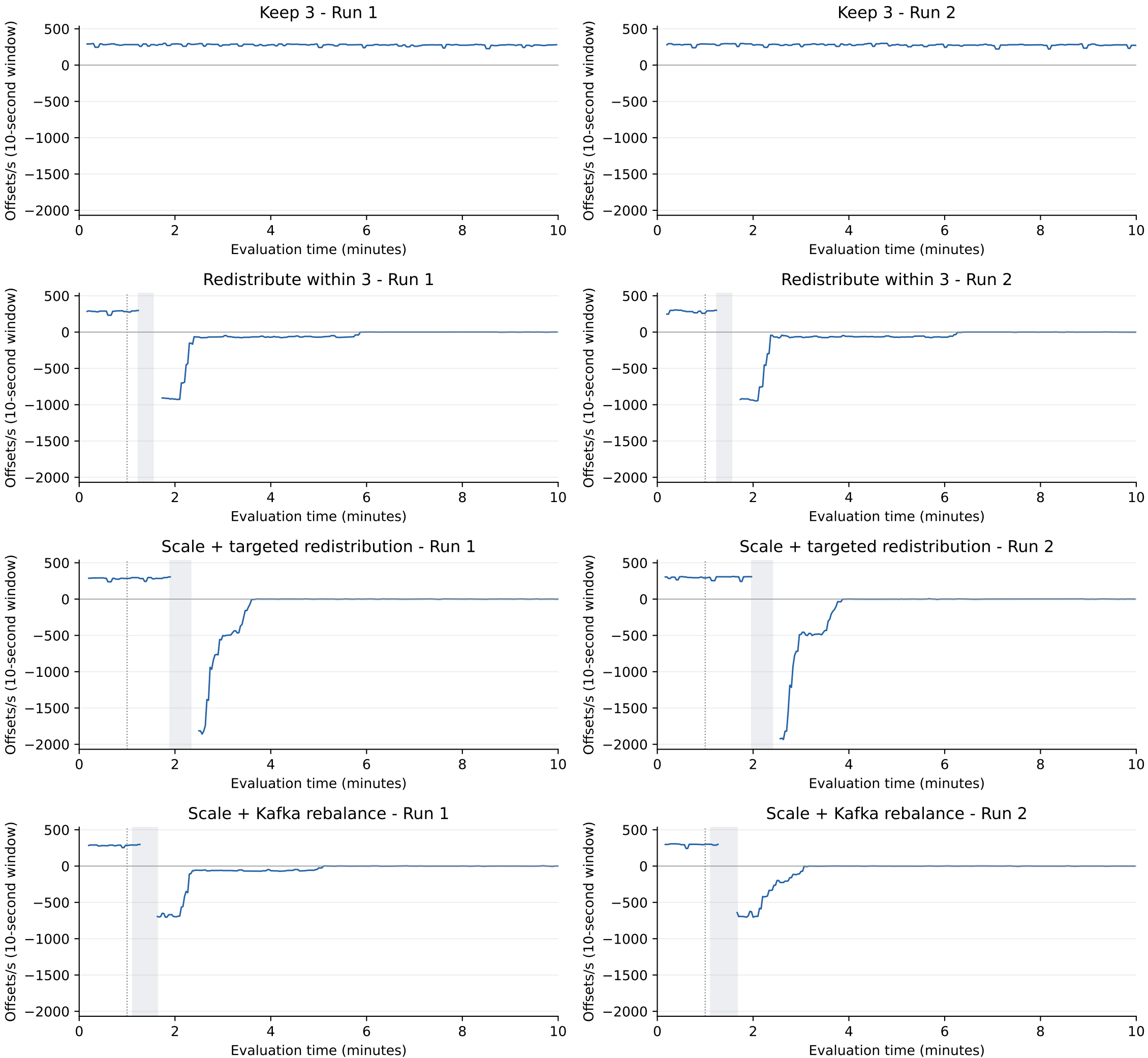


Total lag - 700 messages/s



1 min warm-up + 10 min evaluation + 2 min drain. Common scales; genuine gaps remain unavailable.
Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

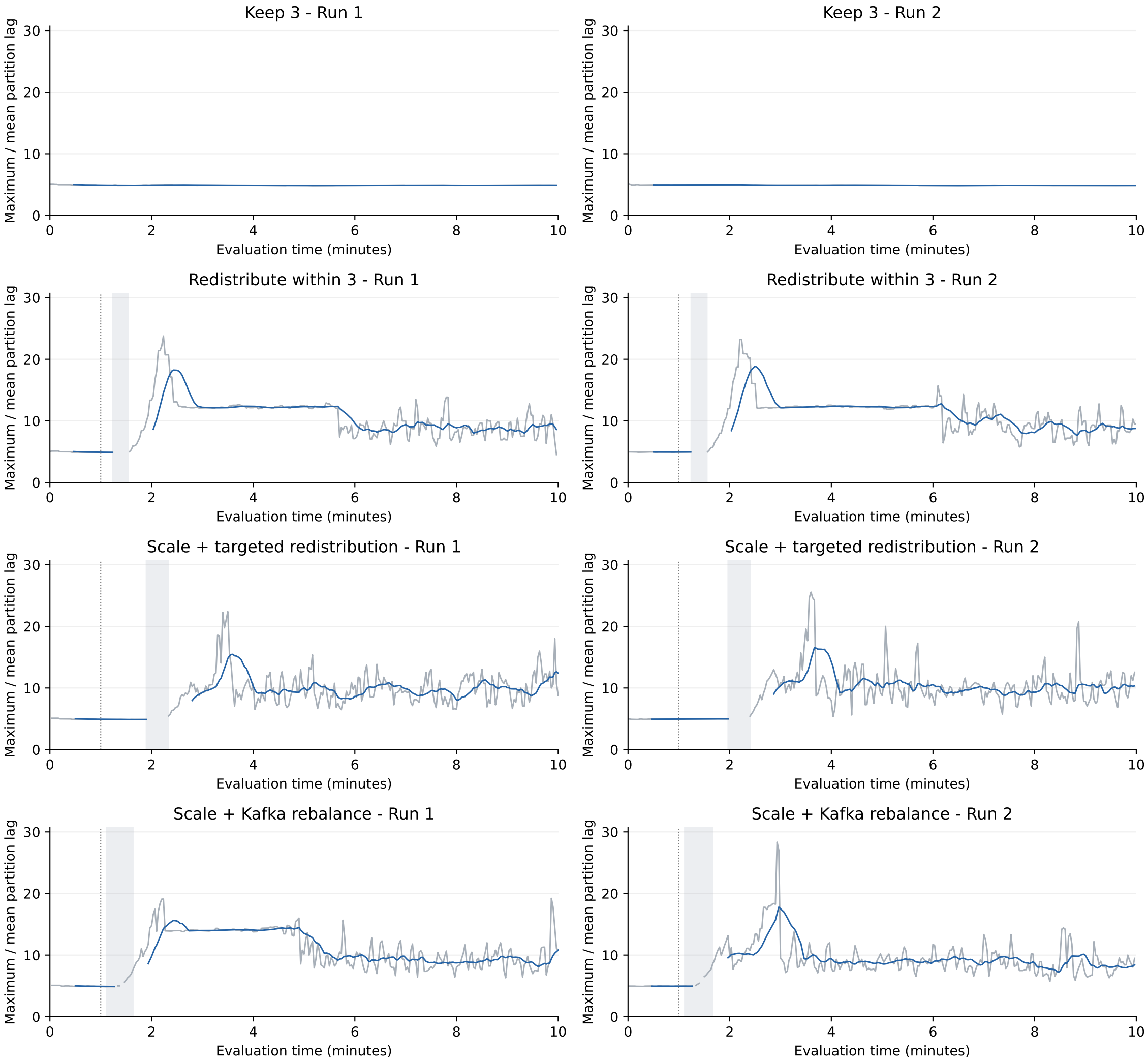
Processing-backlog growth - 700 messages/s



1 min warm-up + 10 min evaluation + 2 min drain. Common scales; genuine gaps remain unavailable.
Growth needs 10 seconds of fresh contiguous observations after a break. Dotted line: scheduled intervention.
Shading: recorded transition interval; this differs from the growth-window waiting time.

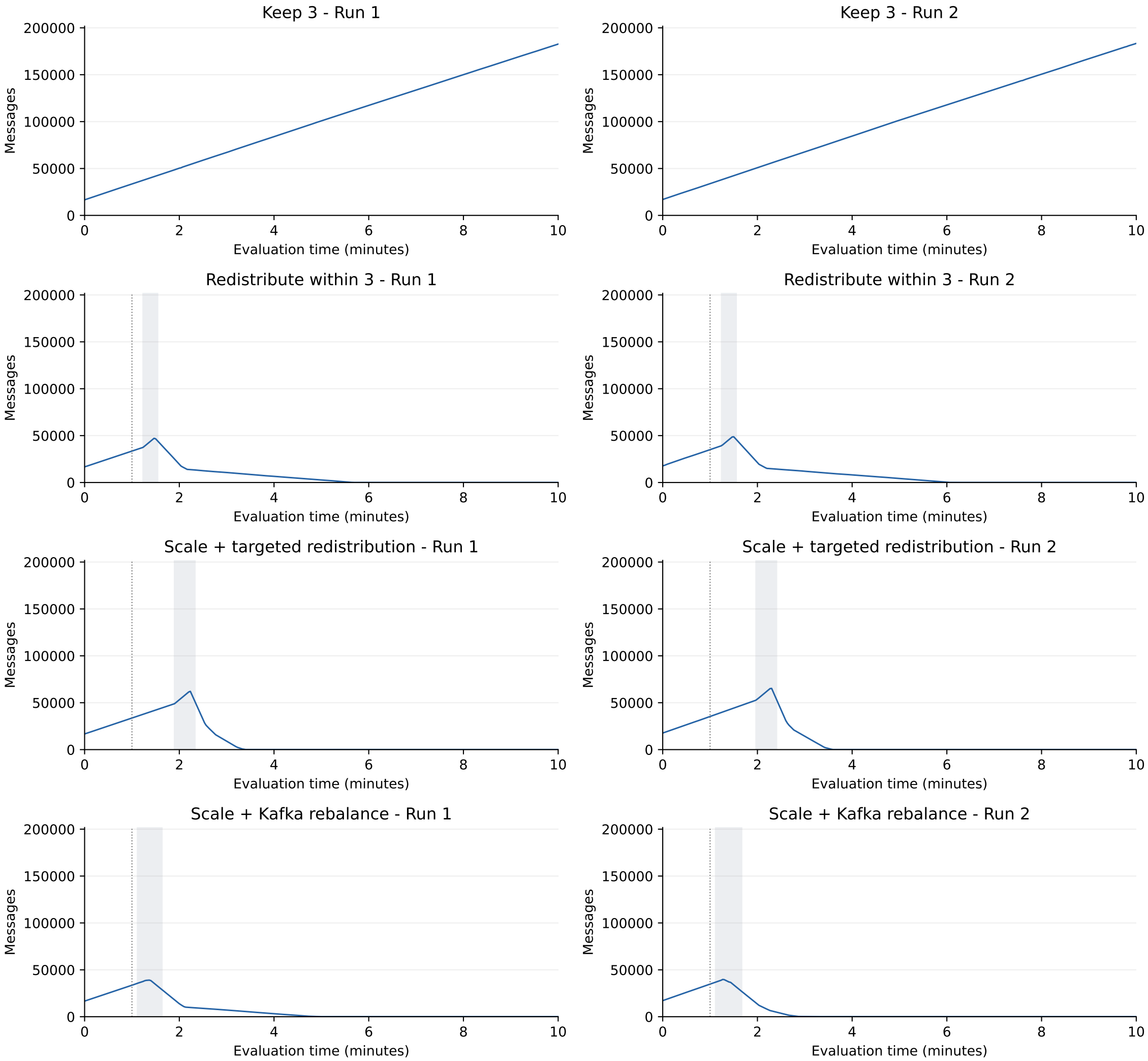
Partition lag skew - 700 messages/s

— Snapshot ratio — 15-sample mean



1 min warm-up + 10 min evaluation + 2 min drain. Common scales; genuine gaps remain unavailable.
Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

Acknowledged but unfinished work - 700 messages/s



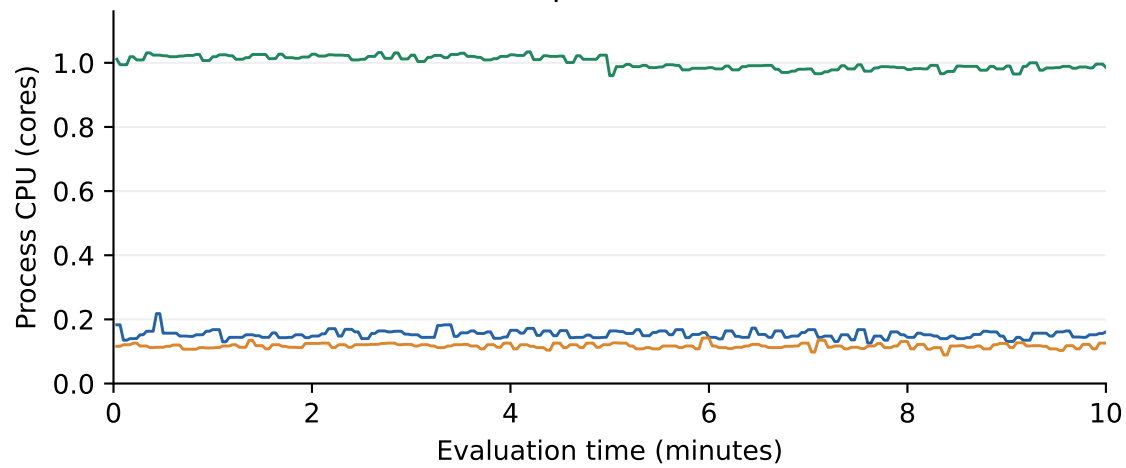
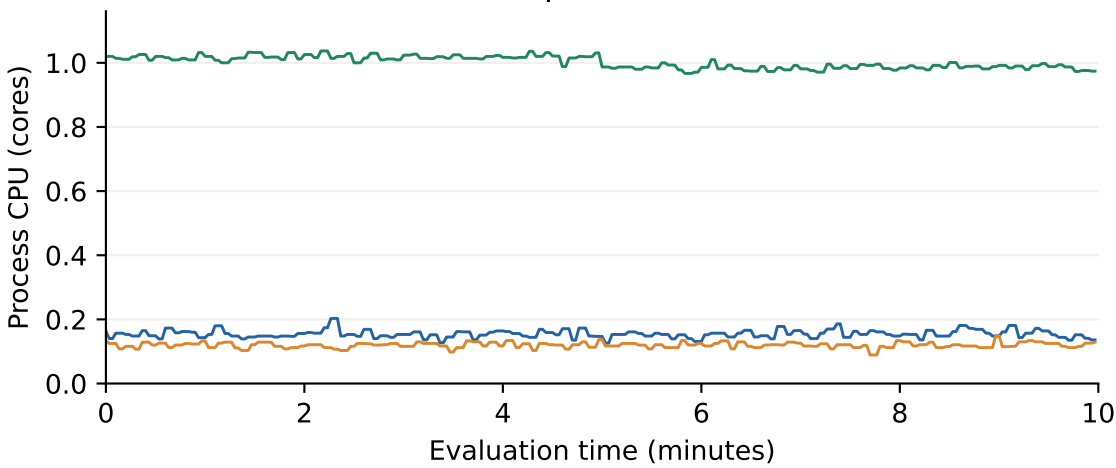
1 min warm-up + 10 min evaluation + 2 min drain. Common scales; genuine gaps remain unavailable.
Distinct acknowledgment/completion events, including warm-up; this is not broker offset lag.

Consumer CPU usage - 700 messages/s

Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5

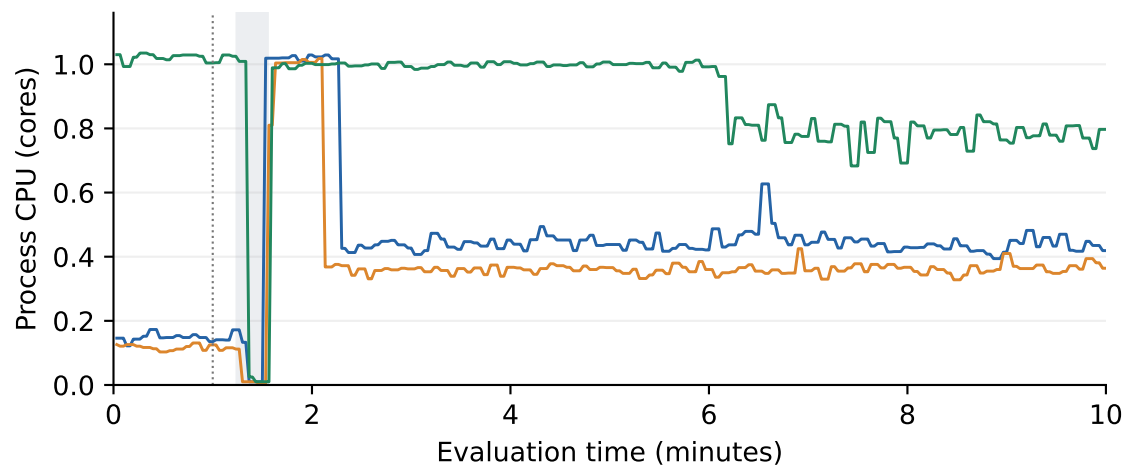
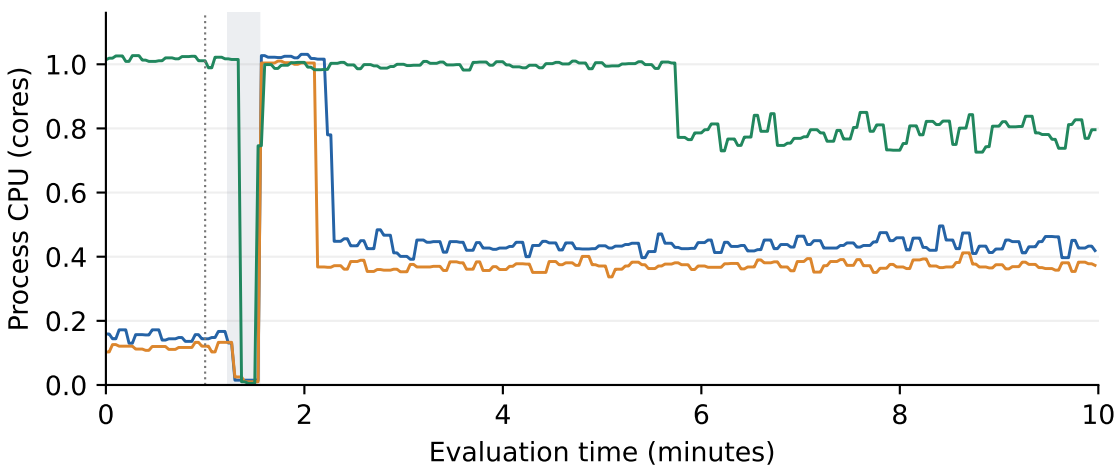
Keep 3 - Run 1

Keep 3 - Run 2



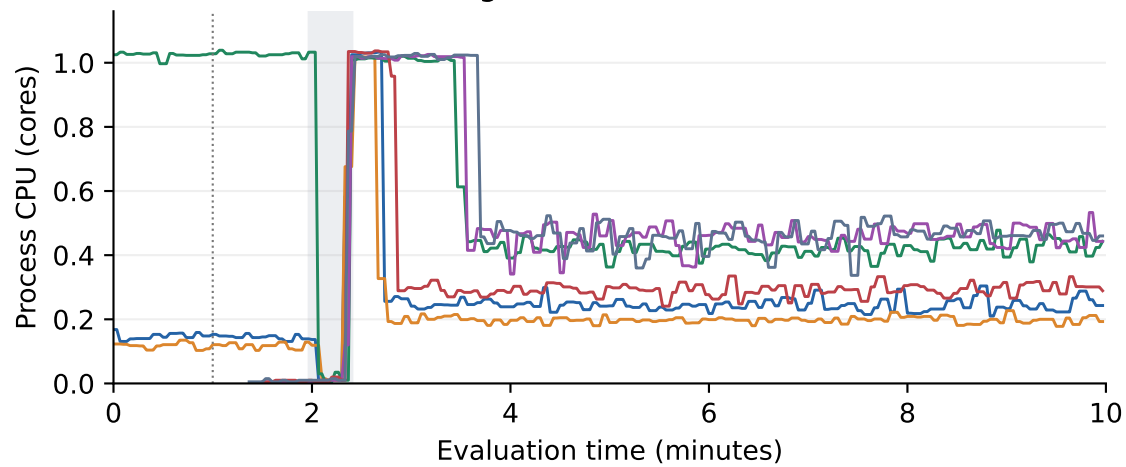
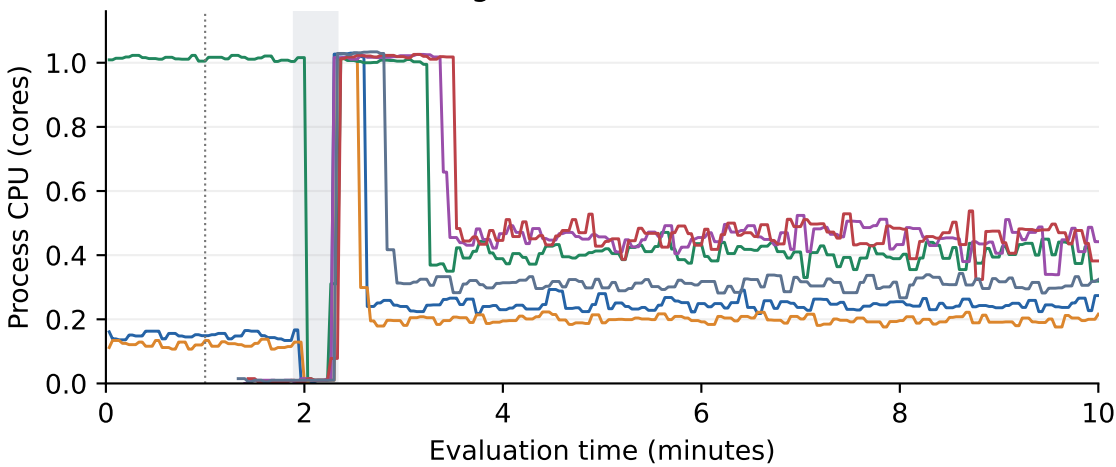
Redistribute within 3 - Run 1

Redistribute within 3 - Run 2



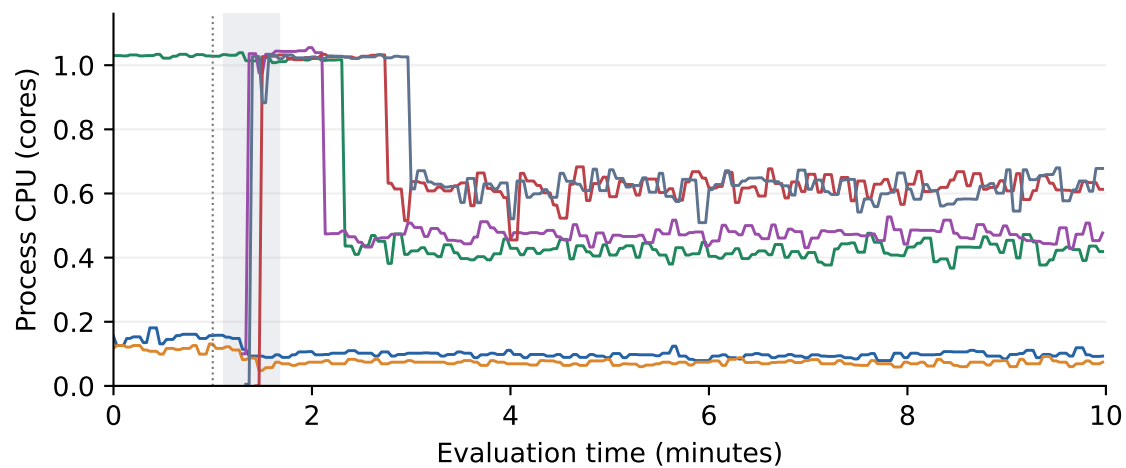
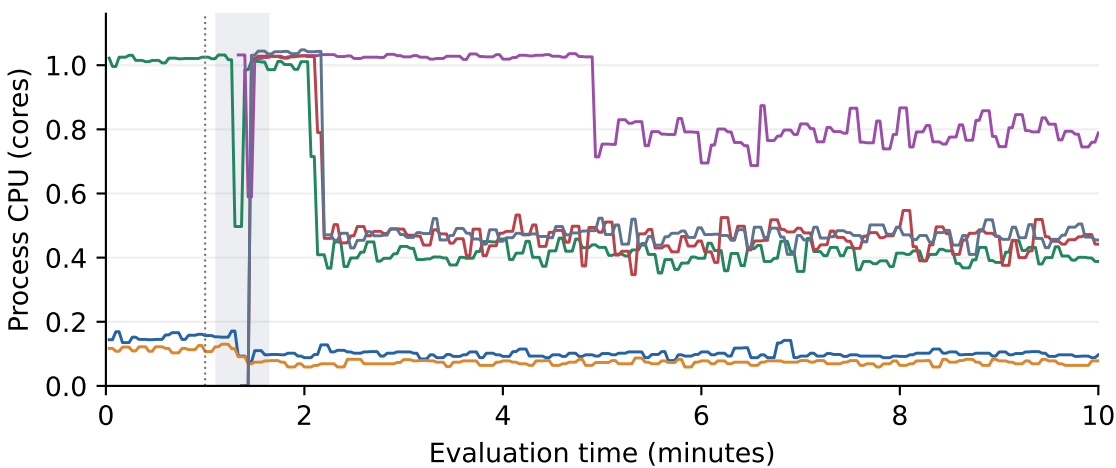
Scale + targeted redistribution - Run 1

Scale + targeted redistribution - Run 2



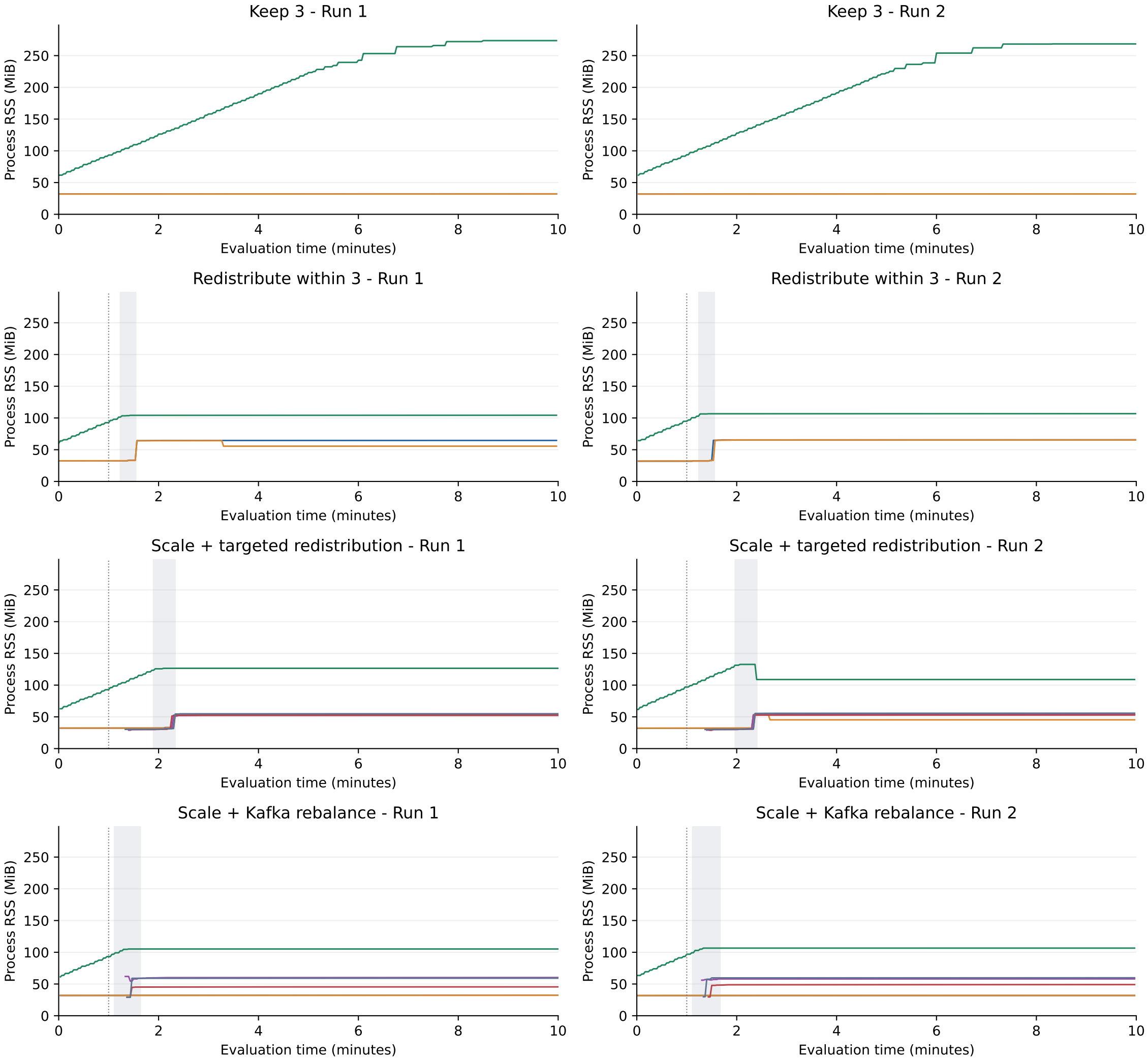
Scale + Kafka rebalance - Run 1

Scale + Kafka rebalance - Run 2



Consumer memory usage - 700 messages/s

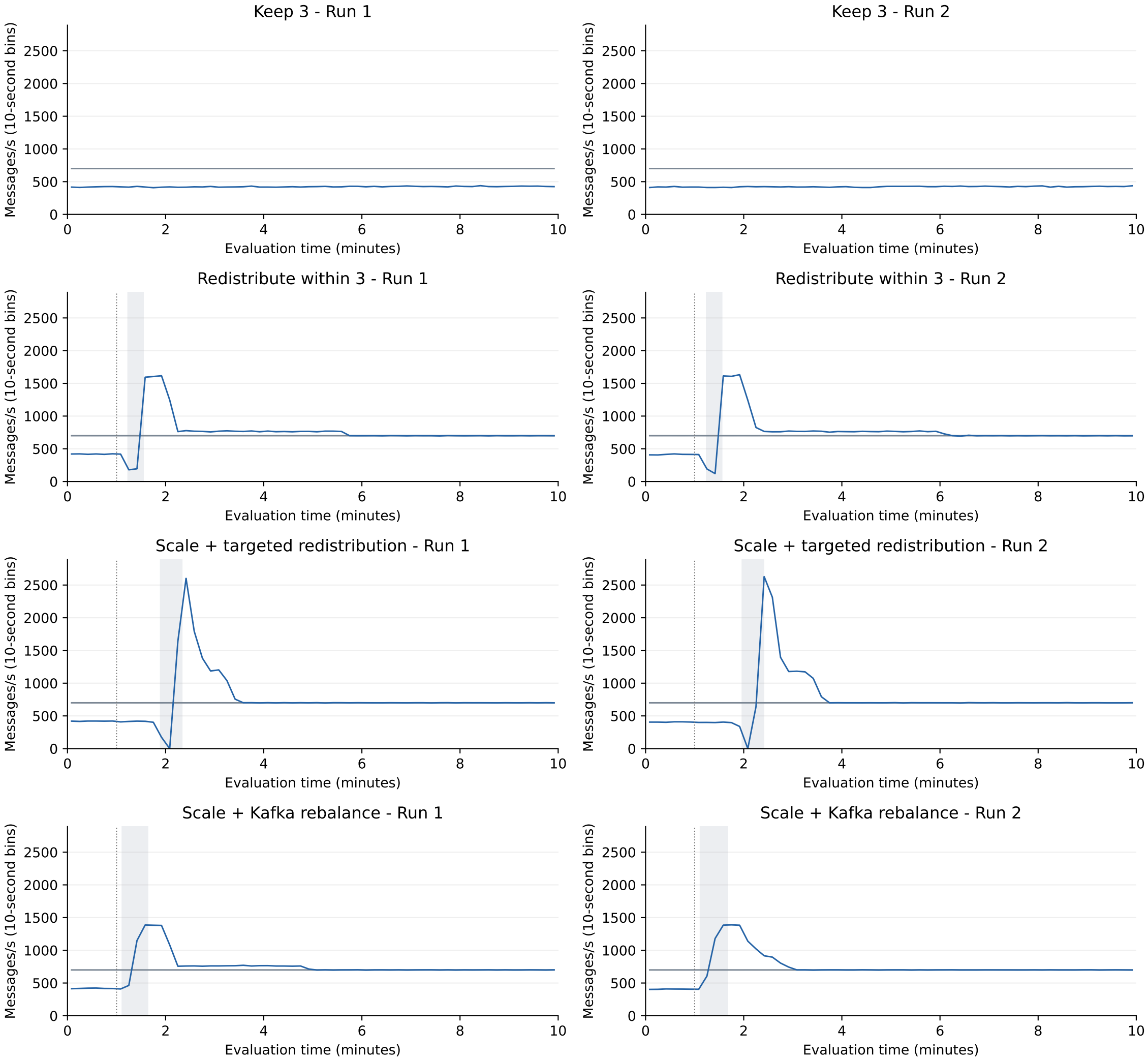
Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5



1 min warm-up + 10 min evaluation + 2 min drain. Common scales; genuine gaps remain unavailable.
Dotted line: scheduled intervention. Shading: recorded transition interval; definitions differ by coordination mechanism.

Input and completion throughput - 700 messages/s

Acknowledged input Unique completions

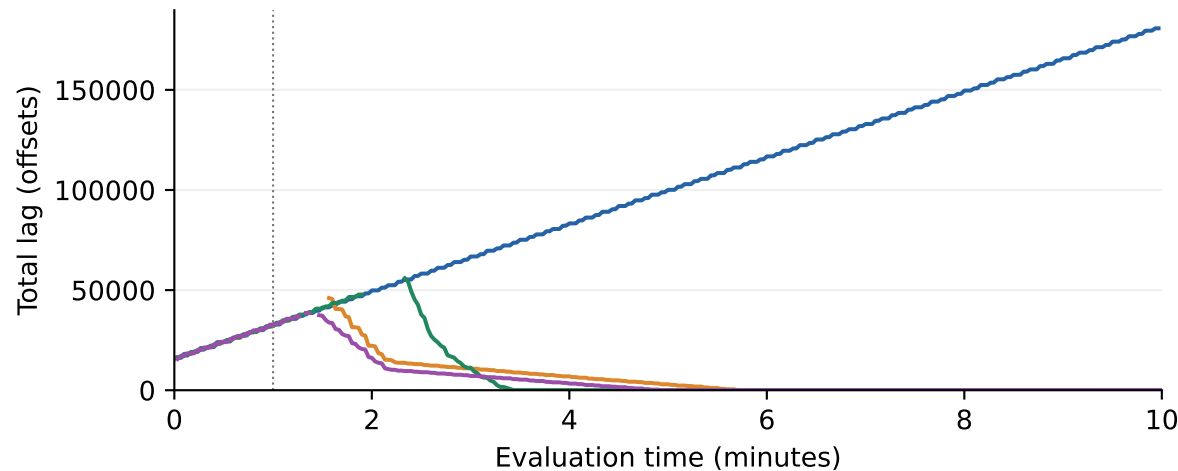


1 min warm-up + 10 min evaluation + 2 min drain. Common scales; genuine gaps remain unavailable.
Completions include warm-up records finishing during evaluation.

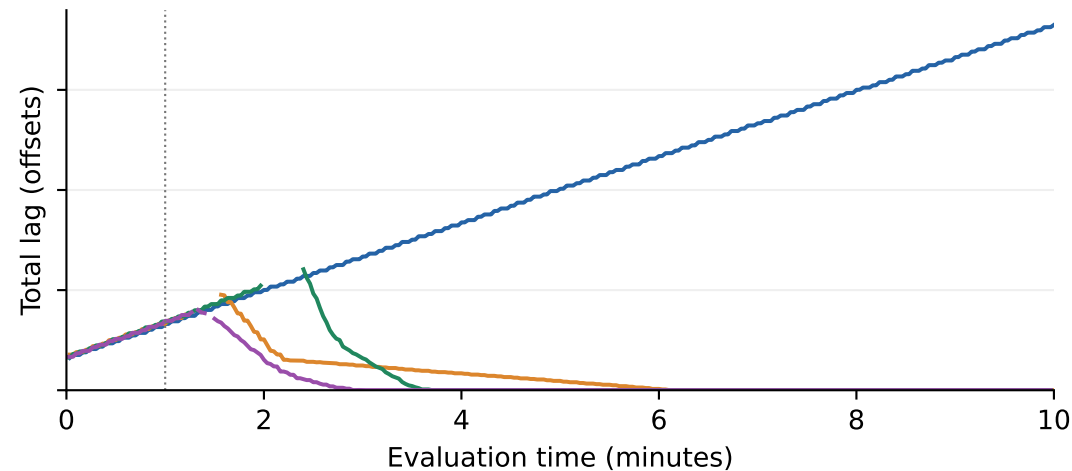
Four responses from the same concentrated ownership - 700 messages/s



Run 1

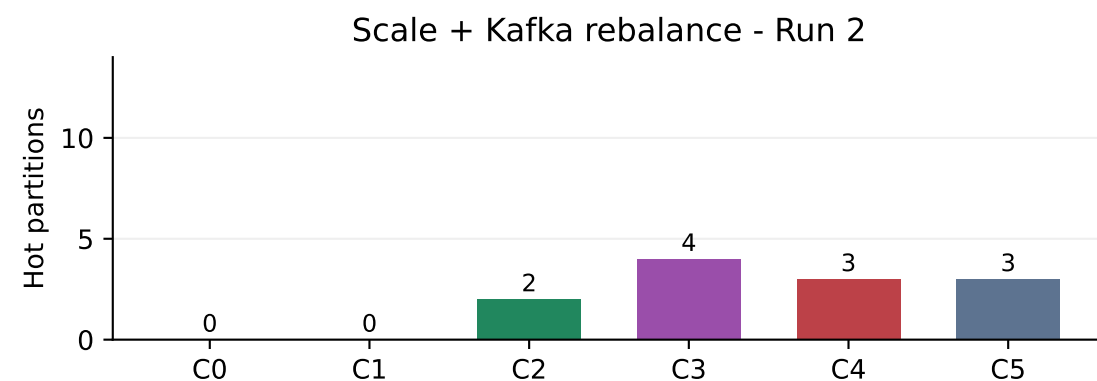
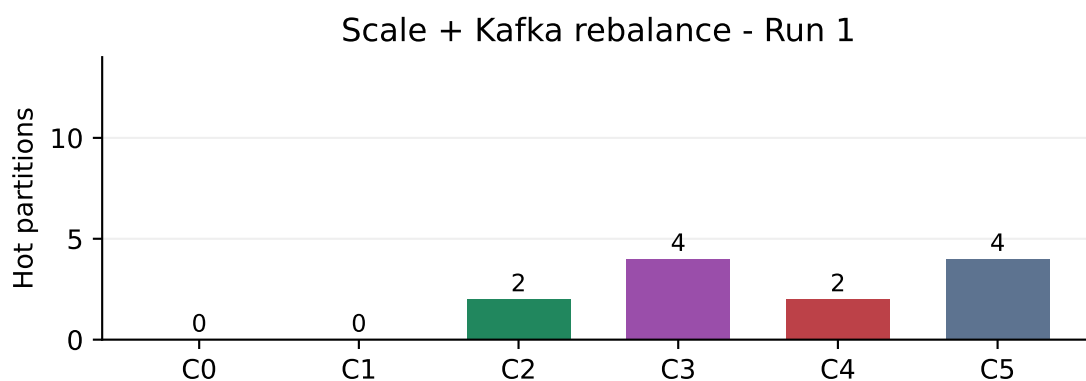
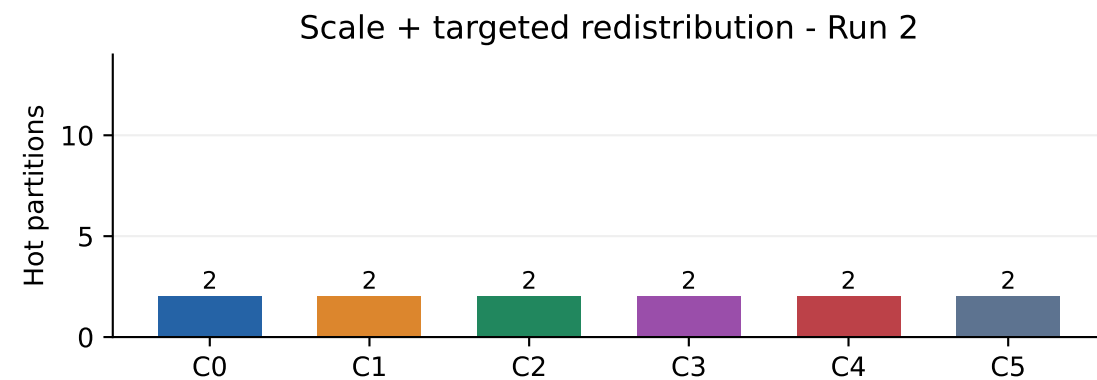
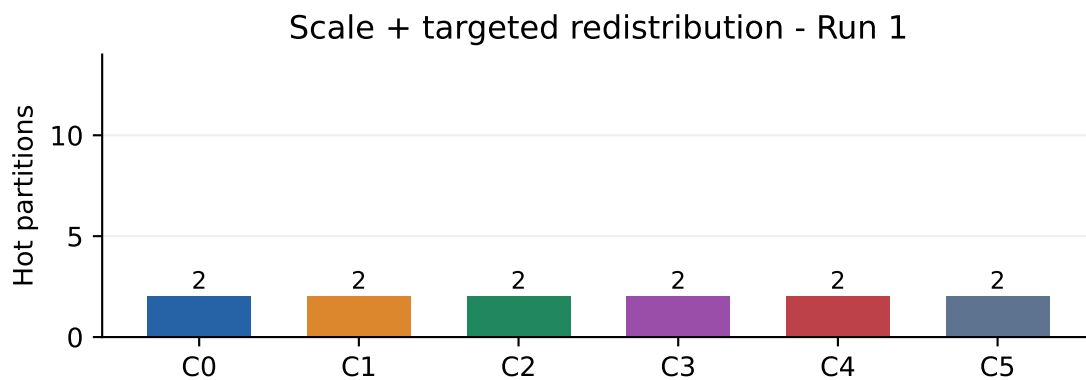
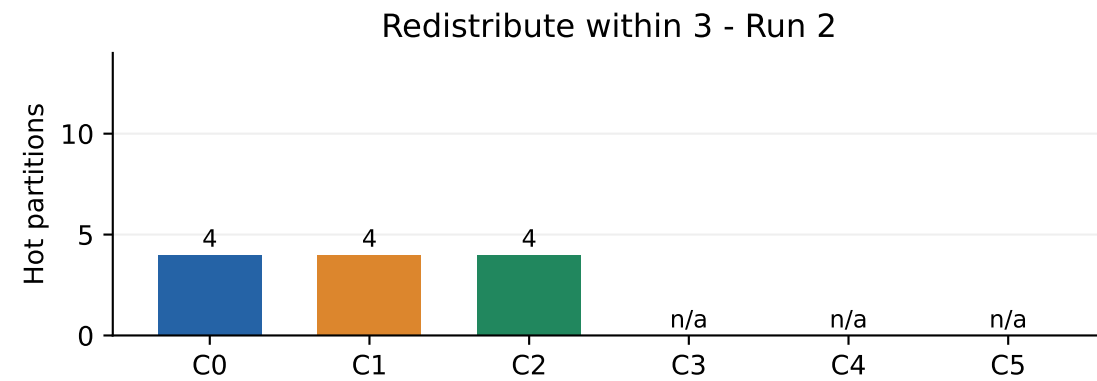
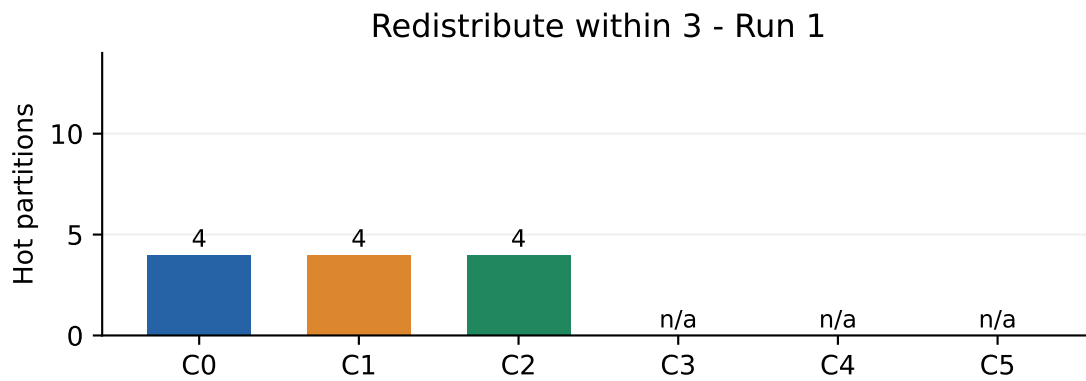
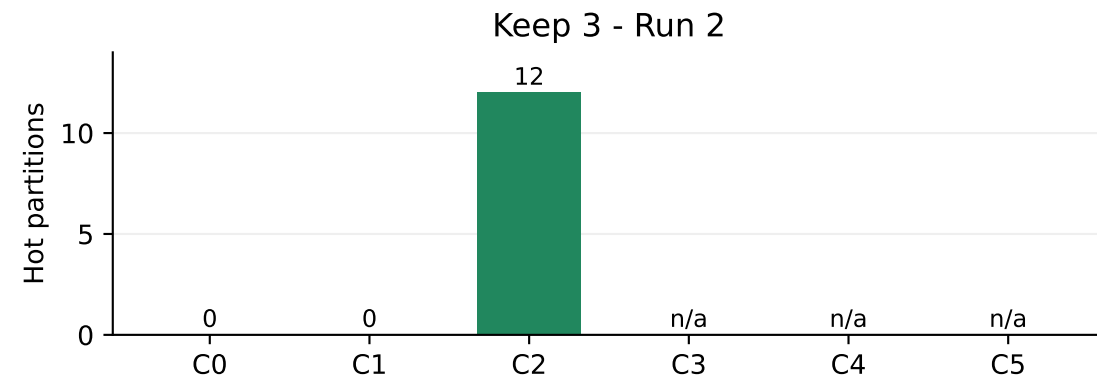
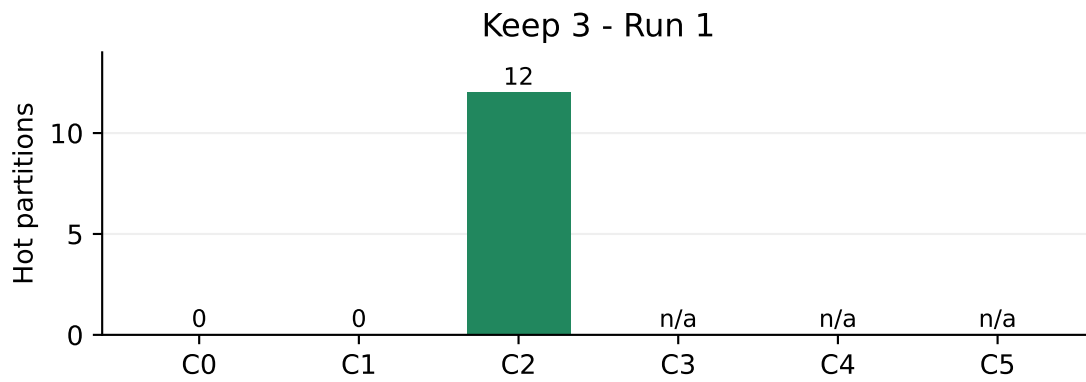


Run 2



Each curve is a separate 13-minute trial; the 10-minute evaluation is shown. Dotted line: scheduled response. Missing observations and ownership changes break curves. Warm-up work remains queued.

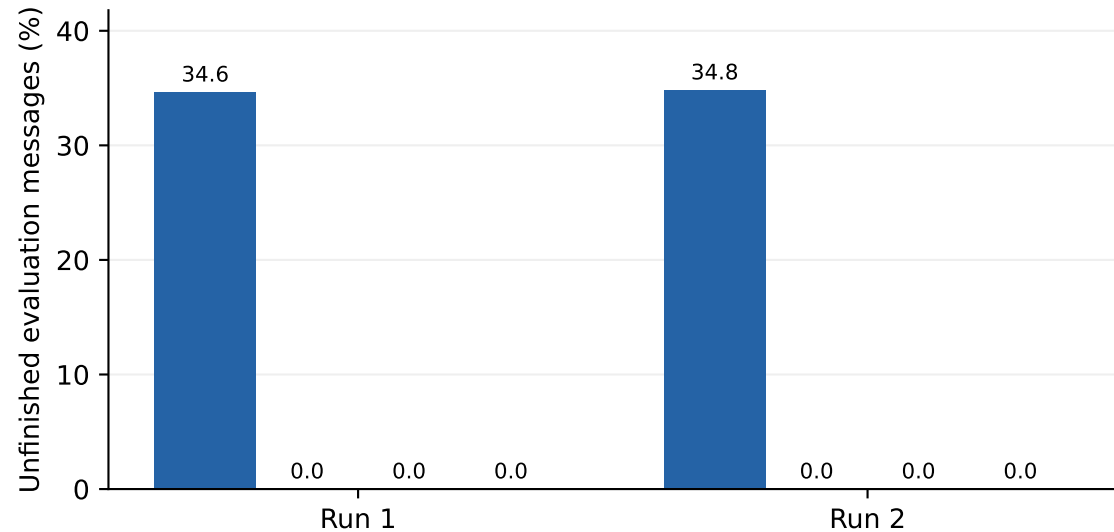
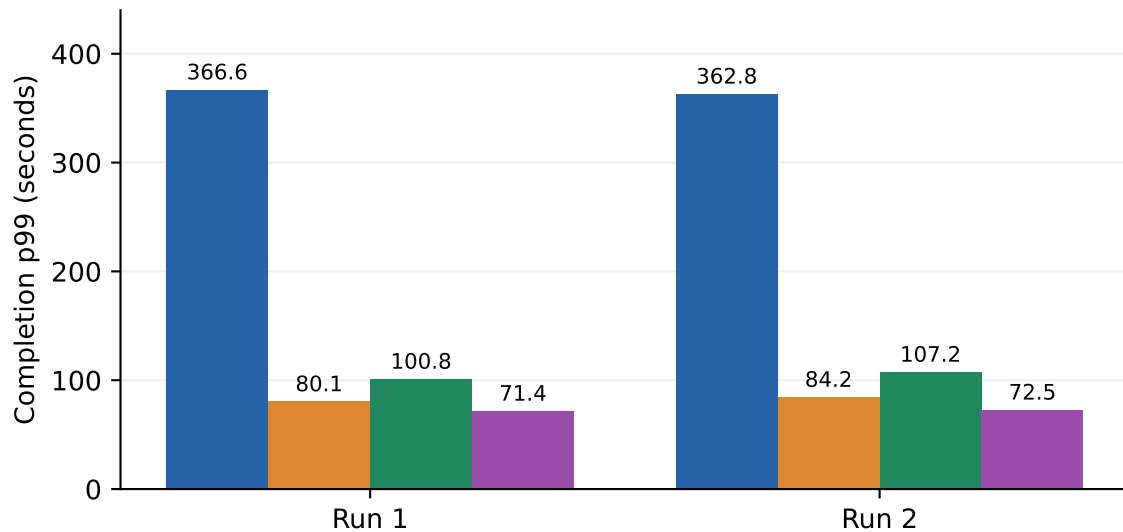
Observed hot-partition ownership at the end of evaluation



All trials started with twelve hot partitions on Consumer 2. C0-C5 identify consumers. Counts use the last complete valid ownership snapshot; n/a means no partition ownership in that snapshot. Equal partition counts do not necessarily imply equal traffic or processing load.

Whole-run completion latency and unfinished work

- Keep 3
- Redistribute within 3
- Scale + targeted redistribution
- Scale + Kafka rebalance



P99 describes completed evaluation messages; unfinished work uses the fixed drain cutoff.

Each trial: 1 min warm-up + 10 min evaluation + 2 min drain. These are separate trials, not within-run before/after p99.